# Research Project Periodic Log Entry Form

01st -15th April

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 1 April | Had a group meeting and discussed the implementation phase after completing the proposal. Planned dataset collection and development tasks. |
| 3 April | Explored open-source datasets on Kaggle and other sources. Found some related tea leaf datasets, but most of them lack proper labeling and tea-specific disease details. |
| 4 April | Filtered and organized collected images into categories such as blister blight, red rust, and grey blight. |
| 6 April | Performed preprocessing steps such as resizing images, removing low-quality samples, and preparing the dataset. |
| 8 April | Explored CNN models like MobileNet and ResNet for disease detection |
| 10 April | Tested a small part of the dataset with an initial CNN model. Observed results and discussed some improvements with the group. |

**Supervisor's Comments**

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:
- Code base checked : YES / NO

Signature : .....

Date : 20/04/2026

# Research Project Periodic Log Entry Form

15th-30 April

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 16 April | Defined binary classification problem (Tea Leaf vs Not Tea Leaf) and organized dataset accordingly for model training. |
| 18 April | Performed preprocessing including resizing images, normalization, and removal of irrelevant samples to improve dataset quality.. |
| 19-20 April | Set up Google Colab environment and prepared dataset for training and testing. |
| 22-24 April | Trained an initial CNN model in Google Colab to classify tea leaf and non-tea leaf images. |
| 26 April | Exported trained model and deployed it using Hugging Face as a backend inference service. |
| 27-28 April | Developed a simple frontend interface and connected it with the deployed model to demonstrate image upload and prediction results. |

**Supervisor's Comments**

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2021

# Research Project Periodic Log Entry Form

01st – 15th May

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 01 May | Organized Tea Leaf and Non-Tea Leaf image dataset for model training. |
| 03 May | Performed preprocessing including resizing, normalization, and augmentation. |
| 06 May | Trained YOLO-based tea disease detection model and tested predictions. |
| 09 May | Connected AI model with backend and frontend interfaces. |
| 12 May | Demonstrated system prototype during PP1 presentation and testing. |

**Supervisor's Comments**

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

# Research Project Periodic Log Entry Form

15th – 30th May

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 15 May | Improved YOLO disease detection accuracy through additional model tuning. |
| 18 May | Added support for multiple tea disease classes and prediction validation. |
| 21 May | Enhanced frontend disease result visualization and confidence display. |
| 25 May | Performed system integration testing with backend APIs and database. |
| 29 May | Fixed prediction issues and optimized model response time for deployment. |

**Supervisor's Comments**

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2020

# Research Project Periodic Log Entry Form

01st – 15th June

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 01 June | Collected additional tea leaf disease samples to improve model robustness. |
| 04 June | Retrained disease detection model using updated dataset and evaluated performance. |
| 07 June | Implemented disease severity analysis and confidence score improvements. |
| 11 June | Connected AI model with backend and frontend interfaces. |
| 14 June | Conducted comprehensive testing and fixed model prediction inconsistencies. |

**Supervisor's Comments**

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

# Research Project Periodic Log Entry Form

16th – 30th June

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 16 June | Improved tea leaf disease classification accuracy through additional model tuning. |
| 19 June | Enhanced disease severity analysis and prediction confidence evaluation. |
| 22 June | Optimized frontend disease result visualization and reporting features. |
| 26 June | Conducted integration testing with backend APIs and resolved prediction inconsistencies. |
| 29 June | Updated model implementation documentation based on testing results. |

**Supervisor's Comments**

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

# Research Project Periodic Log Entry Form

01st – 15th July

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 02 July | Refined YOLO model performance using additional validation samples. |
| 05 July | Improved disease prediction accuracy for multiple tea leaf disease classes. |
| 08 July | Enhanced confidence score presentation and disease result visualization. |
| 11 July | Conducted functional testing of disease detection and prediction workflow. |
| 14 July | Resolved identified prediction issues and optimized model performance. |

**Supervisor's Comments**

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

# Research Project Periodic Log Entry Form

16th – 30th July

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 16 July | Enhanced model evaluation using additional testing datasets. |
| 19 July | Improved backend processing for faster disease prediction responses. |
| 22 July | Conducted end-to-end testing of AI model integration with the web application. |
| 26 July | Optimized disease prediction workflow based on testing observations. |
| 29 July | Updated implementation documentation with recent model improvements. |

**Supervisor's Comments**

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

# Research Project Periodic Log Entry Form

01st – 15th August

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 02 August | Refined disease detection model based on validation results. |
| 05 August | Enhanced disease classification and severity prediction functionalities. |
| 08 August | Improved disease detection dashboard and prediction result presentation. |
| 11 August | Conducted validation testing and resolved identified prediction issues. |
| 14 August | Updated implementation documentation and prepared demonstration materials. |

**Supervisor's Comments**

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

# Research Project Periodic Log Entry Form

16th – 30th August

Project ID : R26-IT-118

Student ID : IT22209320

Name : Miuranga W.A.R

Individual Component : Tea Leaf Disease Detection and Pattern Analysis

**Summary of RP work carried out during the reporting period**

| Date | RP work |
|---|---|
| 16 August | Enhanced the tea leaf disease detection module based on implementation feedback. |
| 19 August | Improved AI model integration with the web application. |
| 22 August | Conducted functional and integration testing of the disease detection module. |
| 26 August | Updated presentation materials and technical documentation for Progress Presentation II (PP2). |
| 28 August | Prepared the tea leaf disease detection module for Progress Presentation II (PP2). |

**Supervisor's Comments**

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026